

REEFPRINT/KHANYA: reading the ore before the mill.

Belt hyperspectral hardness, microscope mineralogy and human-approved plant decisions for South African concentrators

Technical, business and pilot report · Mintek–SCi Grad Hackathon 2026 · Problem 3: Computer vision for real-time mineralogical characterisation

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Abstract

South African concentrators set mill feed, grind and reagents before the laboratory can say what was in the ore. REEFPRINT/KHANYA reads the ore at three speeds and turns the readings into one decision that a person approves. On the belt, a hyperspectral camera feeds a model that predicts grinding hardness (the Bond work index) with a one-sided 90% upper bound. On 146 public drill-core samples the model explained 48% of the variance in hardness (colour alone: 21%), and the bound held for 93% of samples. Replayed as a mill feed-rate policy, it raised simulated throughput by 1.7–1.9% without more overloads. In the laboratory, a microscope model identifies pyrrhotite, chalcopyrite and pentlandite (intersection over union 0.87, 0.58 and 0.55 on 12 held-out sections) and gave no unsafe advice, although those sections also guided development. All results come from public analogue ores and a simulated plant. At Northam’s Zondereinde output, a one-belt pilot costing R8–49 million to install breaks even at 0.05–0.52% more recovered metal after cash costs. We ask TIA, Mintek and one producer for a nine-month pilot; after it, fees are charged only on independently measured value.

Executive summary

The problem	Plants set feed, grind and reagents before laboratory results arrive. South Africa mines about 71% of the world’s platinum, its PGM output fell 9% in 2025, and recovery and mill throughput set revenue.
What we built	A belt hyperspectral hardness model with a safety bound (REEFPRINT), a microscope model for minerals and liberation (KHANYA), and a decision layer in which a person approves every plant change and a signed record keeps it. It runs offline on a laptop.
Evidence so far	Hardness R^2 0.48; bound held for 93% of samples; simulated throughput +1.7–1.9% with no more overloads; microscope IoU 0.87/0.58/0.55 on the three required phases; no unsafe advice on 12 sections (optimistic: the sections guided development). Public data and a simulated plant only: no South African ore has been tested.
What it is worth	At Zondereinde’s output, 1% more recovered metal is worth R44–84 million a year after cash costs (R109–153 million in revenue).
Cost and break-even	Laboratory pilot about R210,000 (TIA). Belt pilot R8.2–49.2 million installed plus R1.6–9.8 million a year, paid at cost by the producer. Break-even: 0.05–0.52% more recovered metal. If the pilot stops at month 9, the most the producer has spent is R9.4–56.6 million.
The ask and the fee	One belt, one truth laboratory, nine months, with go/no-go gates at months 2, 5 and 9. After the pilot the fee is at most 20% of independently measured net value, and nothing if no value is measured.
Main risks	The hardness model may not transfer to Bushveld ore (tested by month 5); the gain may be too small to measure (stop at month 9, no fee).

1 Introduction

As graduate engineers working on Mintek’s Problem 3, our responsibility is to give a concentrator’s shift team a reliable reading of its ore before that ore reaches the mill, and to turn the reading into a plant setting that a person can approve. This report sets out what we built, how we tested it, what it would cost and earn, and how a funded pilot would prove it on South African ore. It is written for metallurgists, mineralogists, process engineers and the people who fund their technology. Technical terms are defined in the glossary (Appendix A).

1.1 Problem statement

The brief states the problem plainly: “without real-time feedback, processing plants cannot adjust to changes in ore qual-

ity, leading to inefficient chemical usage and lower mineral yields” [1]. Ore reaches the mill within minutes of leaving the belt, whereas assays and automated mineralogy are batch laboratory work. Mintek’s own Minerals Processing and Characterisation Cluster runs QEMSCAN automated mineralogy [2], and a commercial laboratory describes mineralogy as “not an overnight assay service” [3]. Consequently, each shift sets the mill and the flotation circuit for ore whose mineralogy it has not yet seen.

The cost of deciding blind falls most heavily on South Africa. Figure 1 illustrates world platinum-group metal (PGM) mine production in 2025, highlighting that South Africa mined an estimated 120 of the world’s 170 tonnes of platinum (about 71%) and 70 of 190 tonnes of palladium. It also holds 63,000 of the world’s more than 76,000 tonnes of PGM reserves, mostly in the Bushveld Complex [4]. The same source attributes the 9% fall in South African PGM output

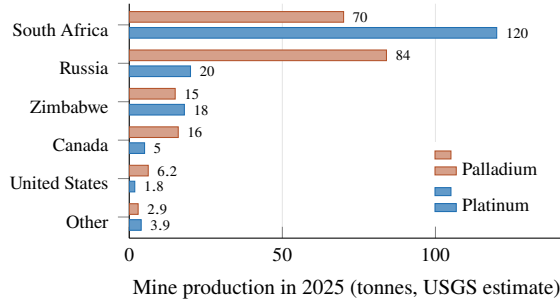


Figure 1: World platinum and palladium mine production in 2025. South Africa mined 120 of 170 tonnes of platinum; Zimbabwe’s Great Dyke ores are of the same family. Data: USGS [4].

in 2025 to lower palladium prices, higher deep-level mining costs and disruptions to the electricity supply.

Recovery is where the money is. UG2 ore carries about 30% Cr_2O_3 against about 0.1% for Merensky ore, and the smelter caps the chromite in concentrate. Mintek testwork made UG2 concentrate at 87% recovery with 2.9% Cr_2O_3 , and recoveries above 90% were possible only if the Cr_2O_3 limit was relaxed [5]. Producers report recovery moves of this size to investors: Valterra raised concentrator recoveries by 1.2–1.7 percentage points at Mototolo and Amandelbult in 2025 [6]. Comminution, which sets both liberation and mill throughput, used 52% of site electricity in a benchmark of mines [7]. Thus, the problem is to read the ore before the mill, state how uncertain the reading is, and turn it into a safe, approved setting at a cost a producer can recover.

1.2 Our solution: three speeds, one decision

REEFPRINT/KHANYA reads the ore at three speeds and turns all three into one decision (Section 3):

1. **the belt** (REEFPRINT), every 30 seconds: a short-wave infrared (SWIR) camera above the mill-feed conveyor feeds a model that predicts each parcel’s grinding hardness with an upper bound, and Bond’s law turns the bound into a mill feed-rate proposal;
2. **the laboratory** (KHANYA), once a shift: a microscope model identifies the sulphide minerals and how well they are liberated, which is the brief’s three-phase deliverable;
3. **the decision**, within the ore’s transit time: an operator approves each change before the ore reaches the mill, or the site’s conservative setting applies, and every step is signed and recorded.

It is important to note what is not claimed: performance on South African ore, a measured plant benefit, mineral identification by the belt camera, or identification of chromite and the silicates by the microscope. Table 1 maps the brief’s requirements and the five judging criteria [8] to the evidence.

2 How the system works

2.1 Data

No South African belt scans or labelled Bushveld micrographs were available, so every model was trained and tested on public data (Table 2). HIDSAG supplies hyperspectral scans of Chilean porphyry copper–molybdenum drill core with laboratory Bond work-index values [9]. LumenStone S2, a Norilsk nickel–copper sulphide ore [10], shares the mineral as-

Table 1: The brief’s requirements and the judging criteria, and where this report evidences each.

Requirement	Evidence (section)
≥ 3 mineral phases	Pyrrhotite, chalcopyrite, pentlandite (4.4)
Accuracy report	Belt R^2 , bound coverage and baselines; microscope per-class, per-section and decision-level results (4.1–4.5, App. B)
Adjust a plant parameter	Mill feed rate from the hardness bound; regrind advice over OPC UA to a simulator (2.3)
Real time	92 ms per parcel; a parcel every 30 s (2.3)
Innovation	A feed-forward ore input that refuses when unsure, with human approval and a signed record (2)
Feasibility	Pricing, support, upkeep, compliance, training, risks (6, 7)
Impact	R44–84 million a year per 1% more metal after cash costs (6.3)
Technical execution	Out-of-fold tests, pass marks set in advance, physics checks (4); 516 automated tests (App. D)
Presentation clarity	Executive summary, glossary, a figure or table per step

Table 2: Datasets. All are public; none was supplied by Mintek or a producer.

Dataset	Ore	Size and use	Licence
HIDSAG [9]	Porphyry Cu–Mo, Chile	146 drill-core composites with Bond work index; visible (39 bands) and SWIR (75 bands). Hardness model	CC0
LumenStone S2 [10]	Ni–Cu sulphide, Norilsk	49 micrographs: 37 train, 12 held-out test. Microscope phases	Research use
Bushveld assays [11]	Bushveld chromitites	1,112 intervals. Check on PGE routing	CC BY 4.0

semblage of Bushveld ores but not their abundance: sulphides cover 62.8% of its sections by area, whereas Merensky ore contains up to about 3% base-metal sulphides and UG2 less [5]. Nevertheless, it is the right teacher for the phases, because the base-metal sulphides in UG2 are predominantly pentlandite, chalcopyrite, pyrrhotite and pyrite [5].

2.2 The belt layer: hardness from the infrared

Sensor (design). The design uses a SWIR line scanner such as the Specim SX25 (960–2500 nm, 640 pixels across, up to 162 lines a second) with a visible–near-infrared (VNIR) companion camera [12]. On a 1.2 m belt moving at 2 m/s it records a line every 12.3 mm with 1.88 mm pixels. It therefore measures parcels of ore, not individual particles. In the pilot design a *parcel* is 30 s of belt (about 60 m, or 4,860 scan lines averaged into one spectrum); in the replay reported here, a parcel is one HIDSAG composite scan of 80×117 pixels.

Model. Each parcel’s spectrum is reduced to features from the infrared absorption regions of clay, mica, chlorite and carbonate minerals (Fig. 4), and a model predicts the Bond work index (W_i , kWh/t). Results are reported only for samples the model never saw (out of fold), and a result counted only if it beat simple baselines under pass marks written down before each run.

Bound and gate. The plant setting uses an upper bound that the true hardness should stay below 9 times out of 10 (a one-sided 90% split-conformal bound [13]), not the best guess. A

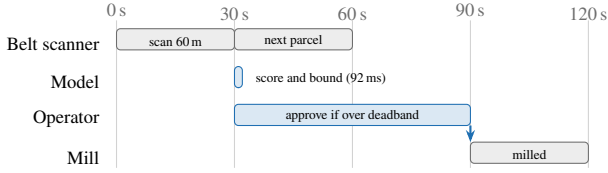


Figure 2: Proposed decision cadence for the pilot, to be agreed with the site. A parcel is scored as soon as it is scanned; the operator then has the rest of the assumed 90 s belt-to-mill transit to approve a change. Once a shift, the metallurgist approves the envelope, ramp limits and deadband. No approval in time, or a flagged parcel, means the conservative setting.

more data-efficient bound [14] was tried first and failed its pass mark, so the exact bound that the plan named as its fallback is used, and the result is provisional. An out-of-distribution (OOD) check flags parcels unlike anything in training: refused parcels receive the conservative setting, and borderline ones the stricter of the bound and the conservative setting.

2.3 The decision layer and its cadence

The decision layer turns the hardness bound into a mill setting that a person can approve. Ball mills are usually limited by motor power, so the tonnes a mill can grind depend on how hard the ore is. Bond’s law gives the specific grinding energy [15]

$$W = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \text{ kWh/t}, \quad (1)$$

where F_{80} and P_{80} are the feed and product sizes (μm) passed by 80% of the material. At a fixed grind target the feed rate the mill can carry is proportional to $1/W_i$, so the proposal is

$$F_{\text{proposed}} = F_{\text{design}} \times \frac{W_{i,\text{design}}}{W_{i,\text{bound}}}. \quad (2)$$

Power, feed size and grind target cancel, so no plant parameter has to be assumed. For example, a design hardness of 16.6 kWh/t and a bound of 16.9 kWh/t give 98.2% of design feed. The proposal must sit inside an operating envelope set by the site’s metallurgist; the demonstration envelope is 85–110% of design feed, with 100% as the conservative setting.

Figure 2 illustrates the proposed cadence, highlighting that a person approves changes, not every computation. A new proposal is computed every 30 s, but the operator sees it only when it differs from the running setpoint by more than a deadband (for example 2% of design feed) set by the metallurgist; smaller changes are not made. The operator may approve, modify within the envelope, reject or escalate. If nobody approves before the parcel reaches the mill (an assumed 90 s), or the parcel is flagged, or the data are stale, the conservative setting applies. The system never holds the last value, because the last value may itself be wrong. Feed may be cut by up to 10% but raised by only 3% per parcel. The shadow phase of the pilot measures how many proposals per hour this produces, and the advisory phase tests whether operators can keep up. Every event is written to a tamper-evident, hash-chained record with conventional (Ed25519) and post-quantum (ML-DSA-65 [16]) signatures. The microscope’s advice reaches a simulated regrind setting over OPC UA, the industrial data-exchange standard [17]; the same interface is designed for the feed rate.

2.4 The laboratory layer: minerals under the microscope

The microscope model is a standard image-segmentation network (DeepLabV3 with a ResNet-50 backbone [18, 19]) that labels every pixel of a reflected-light image as resin, pyrrhotite, chalcopyrite, pentlandite or magnetite. Only the approved checkpoint (SHA-256 de7135a9, full hash in Appendix D) may send advice to the simulator. A *Quick* scan classifies six 512-pixel fields of a section; a *full* scan covers the whole section.

Connected regions of the mask are split into grains. Chalcopyrite and pentlandite are the valuable (payload) phases, and pyrrhotite is rejected. A grain is *free* if at least half its area is payload and *locked* otherwise; the 50% line is the boundary between low and high middlings in the liberation classes used by Mintek’s characterisation group [2]. A rule-based advisor turns these measures into *grind finer*, *continue* or *verify before acting*. It answers *verify* when the liberation estimate lies within ± 0.335 of the 50% line, and it gives no recommendation with fewer than 9 payload-bearing grains or mean confidence below 0.85. The 50% line is sourced; the ± 0.335 band was calibrated on the 12 test sections (Section 4.5); the other thresholds are placeholders for site metallurgists to set.

2.5 How we tested it

Belt models were scored out of fold against the strongest simple baseline (the fold mean, the fold median and a colour-only model), with pass marks committed to the repository before each run. The microscope was scored on the publisher’s 12 held-out S2 sections. These sections were never used to train the network, but they were evaluated many times during development: to compare pipelines, to choose the approved checkpoint over a retrained candidate, and to calibrate the advisor’s uncertainty band. Microscope results are therefore optimistic, and an independent, locality-disjoint test set is the first deliverable of the laboratory pilot. Accuracy is the intersection over union (IoU: the overlap between predicted and expert-labelled areas, from 0 to 1), pooled over 103.8 million pixels and averaged per section, with bootstrap confidence intervals over sections. Decision-level agreement runs the advisor on the model’s mask and on the expert’s mask; a disagreement is *unsafe* if the model gives a confident instruction that the expert mask does not support, for example *continue* where the expert mask says *grind finer*.

3 System architecture

Figure 3 illustrates the system, highlighting that both layers pass the same safety gates and the same human decision. The belt monitor (REEFPRINT Live) runs the hardness model, the decision view and the signed record. The microscope workbench (KHANYA) is a web application (React) on a Python service (FastAPI and PyTorch) that runs on an ordinary laptop processor. In local mode it accepts connections only from the same machine and makes no external requests, so a network outage at the plant does not stop the advice. No language model computes a measured value or a control value.

Speeds were measured on a four-core laptop processor: 92 ms per parcel (median); 16.7–21.2 s per six-field microscope scan on mains power (5 runs) and 30.8 s on battery.

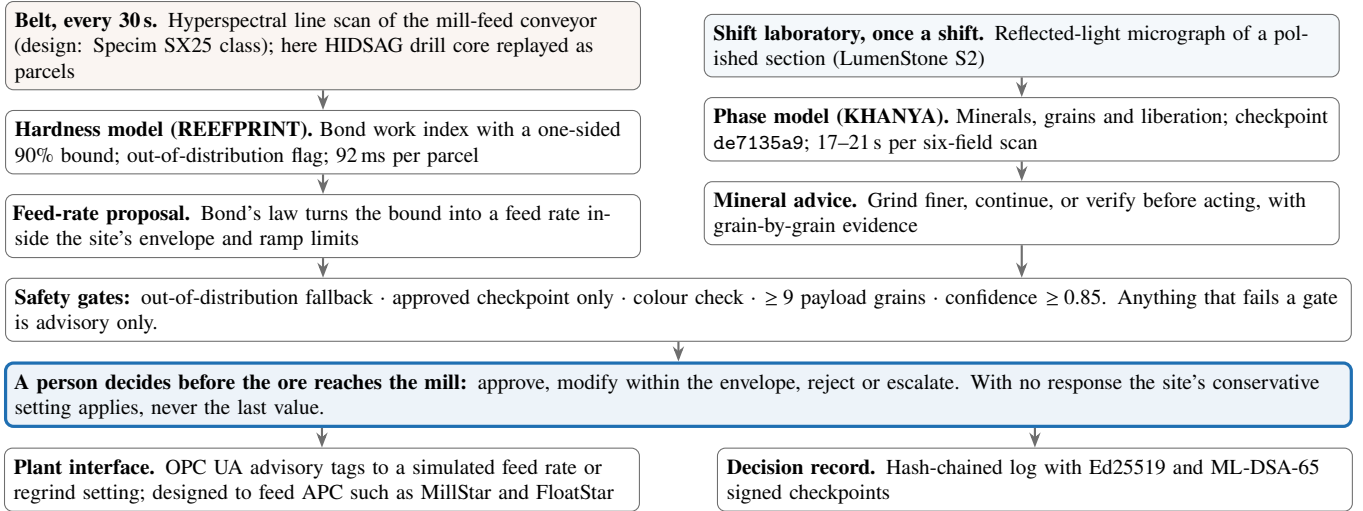


Figure 3: System architecture: three speeds, one decision. The belt layer (left) is the site system; the microscope layer (right) is the laboratory truth layer and delivers the brief’s three phases. Both pass the same gates and the same human decision. Everything runs offline on one laptop processor; the plant is a simulator.

A static public demonstration runs the belt monitor on pre-computed data; a secure server passed 17 of 17 security tests locally and is not deployed. Software verification is in Appendix D.

4 Results

4.1 Belt: predicting grinding hardness

Of the five HIDSAG targets tested (copper and molybdenum recovery, lime consumption, pH and Bond work index), only the work index beat its strongest baseline under every pass mark. Figure 5 illustrates the out-of-fold results, highlighting that the infrared carries most of the information: the hyperspectral model explains 48% of the variance in work index, a model given only colour explains 21%, and the average of the training data explains nothing.

Figure 6 illustrates how the safety bound behaves, highlighting that it was exceeded on 10 of 146 samples (6.8%), within its 10% design rate. The price of that safety is width: the bound sits on average 1.64 kWh/t, about 11% of the mean hardness, above the best guess. The OOD check passed 137 parcels, marked 5 borderline and refused 4. Figure 4 shows one parcel as the model sees it. For parcel GMET-0004 the predicted W_i is 15.9 kWh/t and the bound 16.9 kWh/t, against a design hardness of 16.6 kWh/t (the training data’s 90th percentile), so the proposal is 98.2% of design feed. The laboratory value was 12.1 kWh/t: the bound was conservative, which is the direction a safety bound should err.

4.2 Belt: from hardness to a plant decision

Figure 7 shows the decision for the same parcel as the operator sees it. Figure 8 illustrates four feed-rate policies replayed over all 146 composites, highlighting the trade-off between throughput and overloads. Setting the feed from the bound, with the OOD fallback, gave a simulated throughput gain of +1.9% (95% CI +0.9 to +2.9%), and overloaded parcels fell from 8.9% to 6.8%. The one-sided 95% upper bound on that change (+1.4 points) is inside the +3-point margin set before the test. Held inside the 85–110% envelope, which the pre-registered replay did not apply, the gain is +1.7% (+0.7 to

+2.7%) with the same overload share. Feeding from the best guess instead would add 13.6% but overload 45.2% of parcels, which is why the policy uses a bound. Perfect knowledge would add 14.0%, so the bound captures about one-seventh of what is available.

The result is provisional for three reasons: the exact bound was adopted after the more efficient one failed, HIDSAG has no time order (so transit time, stockpile mixing and the asymmetric ramp limit are not simulated), and the ore is not Bushveld ore. Symmetric ramp limits were tested over 200 random parcel orders: $\pm 10\%$ per parcel left overloads unchanged (-0.3 points), but $\pm 5\%$ added 2.1 points and $\pm 2\%$ added 4.7, so feed must be cut fast. A pre-registered robustness test found that a new capture of the same ore keeps its error ($\times 1.00$), as do half the field of view, added noise and $\pm 15\%$ illumination; a wavelength drift raised the error by a factor of 1.25 unflagged, so a wavelength-calibration check was added to the design.

4.3 Is it physically possible?

Every claim was tested against its physics, and the failures stay in the report. Bond’s law gives the throughput gain above (provisional). Grinding faster costs at most 0.25 percentage points of flotation recovery on the steep part of the recovery curve and none on the plateau (modelled), so residence time must be checked first. Re-ordering ore during load-shedding cannot create tonnes, because energy per tonne is conserved over a cycle: it gave -3.8% in simulation, and even perfect knowledge gave -3.0% , so that claim was rejected. The camera’s 12.3 mm lines and 1.88 mm pixels resolve parcels, not particles; no test showed that the belt camera can identify minerals, so the microscope carries the phases; and the system is optical only, with no radioactive source and no radiation sensing.

4.4 Laboratory: phase identification

The microscope meets the brief’s three-phase requirement. Figure 9 illustrates its accuracy by class, highlighting IoU values of 0.87 for pyrrhotite, 0.58 for chalcopyrite and 0.55 for pentlandite (precision and recall in Appendix B). The pooled mean IoU is 0.5725 (95% CI 0.494–0.624); averaged per sec-

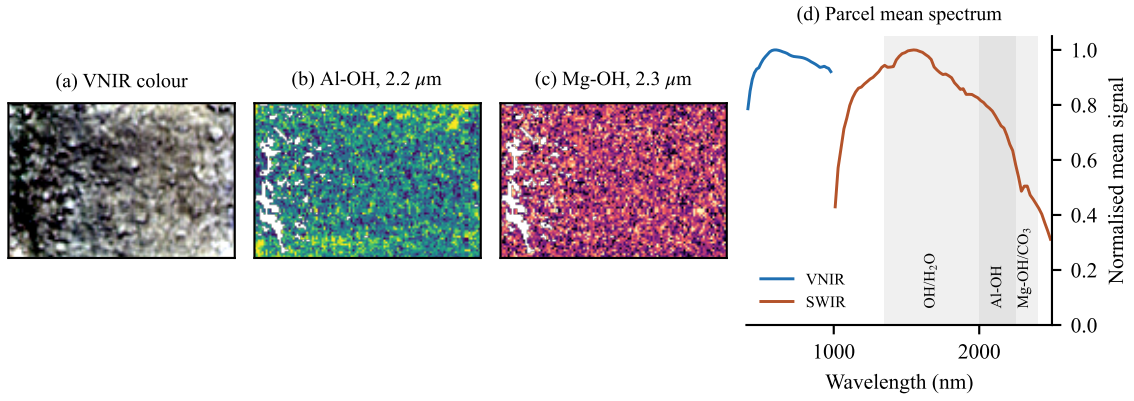


Figure 4: One belt parcel as the hardness model sees it: HIDSAG composite GMET-0004 (Chilean porphyry Cu–Mo), 80×117 pixels. (a) Colour image from the visible bands. (b, c) Relative absorption depth at the Al–OH (white mica, clays) and Mg–OH (chlorite, biotite, carbonate) features; detector stripes removed for display only. (d) Mean spectrum, with the absorption regions the model uses. The model predicts W_i 15.9 kWh/t with a 90% upper bound of 16.9 kWh/t (laboratory value 12.1 kWh/t). The maps show absorption features, not a mineral identification.

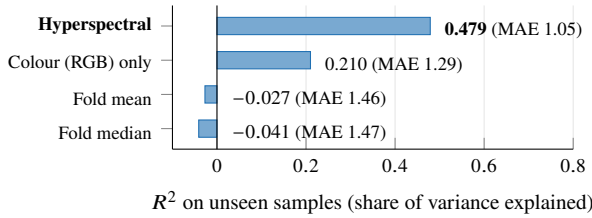


Figure 5: Bond work index on 146 HIDSAG composites, out of fold; mean absolute error (MAE) in kWh/t. Mean measured W_i is 14.62 kWh/t.

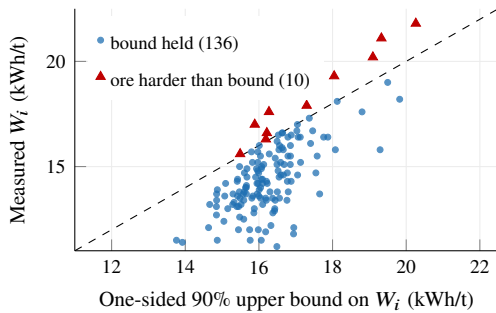


Figure 6: Calibration of the hardness bound, out of fold. Points below the dashed line are parcels the bound covered: 93.2% (95% CI 88.4–96.6%) against the 90% target. The bound sits on average 1.64 kWh/t (range 1.04–2.14) above the prediction.

tion it is 0.4671 (0.412–0.534). The model clearly beats trivial baselines (majority class 0.117; colour-only 0.395).

Magnetite, a fourth class the brief does not require, is never predicted. It is only 0.8% of the test pixels: 78% of it is labelled as resin, 17% as pyrrhotite and 5% as pentlandite. The last share is about 0.3% of all pixels predicted as pentlandite, so magnetite barely moves the payload measures; where it is intergrown with a sulphide grain, dropping it slightly overstates that grain’s payload share.

A retrained candidate (42646cfa) scores 0.632 overall and does detect magnetite (IoU 0.25), mainly because chalcopyrite improves (0.800 against 0.576). However, it loses pyrrhotite (0.818 against 0.870) and pentlandite (0.446 against 0.547), and only half of the pixels it calls pentlandite are pentlandite (precision 0.50 against 0.68). Pentlandite is the main nickel and PGM carrier, and false pentlandite inflates the payload

Table 3: Decision-level confusion on the 12 held-out sections: advice from the expert mask (rows) against advice from the model’s mask (columns).

Expert mask \ model	Grind	Continue	Verify	Too few	Total
Grind finer	3	0	3	2	8
Continue	0	1	2	0	3
Adjust reagent	0	0	1	0	1
Total	3	1	6	2	12

share, pushing advice toward *continue*, the unsafe direction. The approved checkpoint is therefore the safer one. The candidate has not been evaluated at decision level, and a second training run of the approved recipe scored 0.454 because the patch sampler was not seeded; the sampler is now seeded (Appendix D).

4.5 Laboratory: decisions and grain evidence

Table 3 compares the advice from the model’s masks with the advice from the expert masks. The model gave a definite instruction on 4 of 12 sections, all correct, and hedged on the other 8: verify on 6, and no recommendation on 2 sections with too few payload grains. None was unsafe; with 0 of 12 the one-sided 95% upper bound on the unsafe rate is 22.1%. On the expert masks the advisor always gives a definite answer, so the hedges come from model error. The advisor is therefore safe but often non-committal.

It is important to note that the ± 0.335 verify band was calibrated on these same sections, so the safety result is partly in-sample. Replaying the liberation rule on the stored section measurements shows how much it depends on the band: on the 9 sections it governs, the result stays at 0 wrong for bands of ± 0.25 or wider (7 correct at ± 0.25), but one section (test_04) becomes unsafe below that, and two with no band at all. The band must therefore be re-set on independent sections in the laboratory pilot. A realistic exposure shift (–35 levels on each colour channel) changed the advice on 8 of 12 sections, so illumination control is also a pilot requirement.

Figure 10 illustrates the grain inspector, highlighting how a mineralogist can audit the advice grain by grain. In Quick mode each field is classified without context beyond its edge,

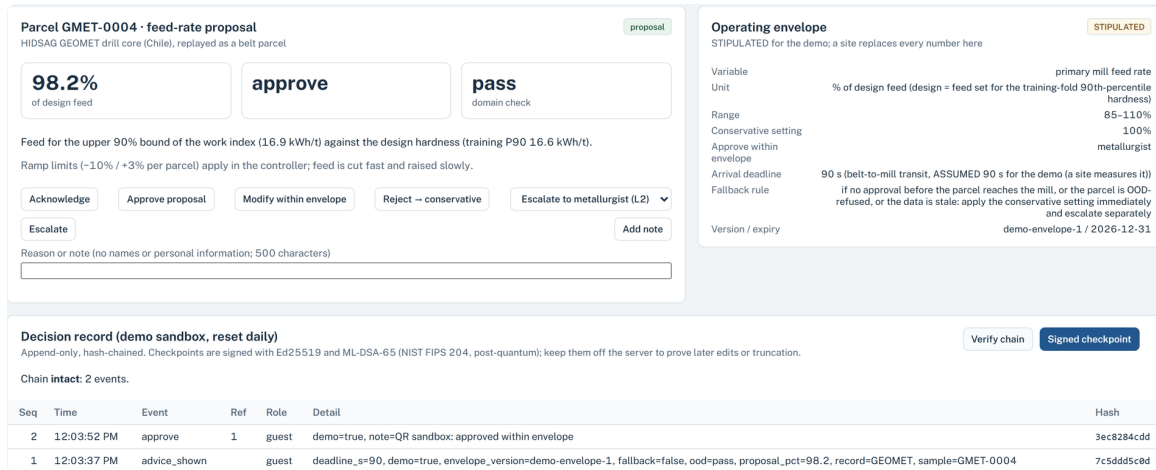


Figure 7: The Decisions view of REEFPRINT Live for parcel GMET-0004: a feed-rate proposal of 98.2% of design inside a demonstration envelope, the operator's actions, and the hash-chained decision record with a guest approval (demonstration sandbox). Screenshot of the application.

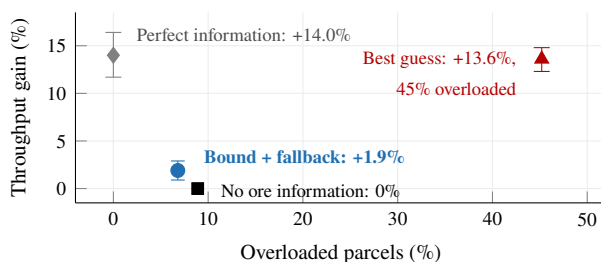


Figure 8: Feed-rate policies replayed over 146 composites (simulated; not PGM ore and not a time series), with 95% confidence intervals. Up is more throughput; left is fewer overloads.

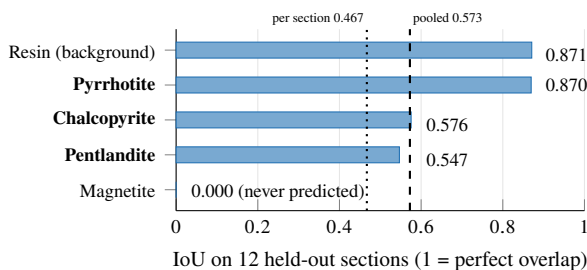


Figure 9: Microscope accuracy by class for the approved checkpoint, full-section inference. The three phases the brief requires are in bold. Section means range from 0.340 to 0.728 (Appendix B).

and on test_11, 19 of 24 grains come within 6 pixels of a field edge. Thus, Quick-mode grain statistics are indicative, and the full scan is the reference.

5 Validation against literature and limitations

5.1 Validation against literature

The results align with published work and with what producers report. Notole et al. [20] validated hyperspectral imaging and portable XRF against reference methods on Merensky Reef drill core scanned at the Council for Geoscience in Pretoria, so the sensing layer has been shown to work on Bushveld material, although not yet for hardness. Our 50% free/locked line sits on the boundary between low and high middlings in Mintek's liberation classes [2], and the three sulphides the mi-



Figure 10: The grain inspector on section test_11 (Quick scan): one analysed field with the phase overlay (left; teal pyrrhotite, violet pentlandite) and the selected grain's evidence (right). Grain 30 (circled) is 76.5% pyrrhotite and 23.5% pentlandite, so it is locked; grain 31, the violet grain to its right, is pure pentlandite and free. Cropped from a screenshot of the application.

croscopie identifies are the main base-metal sulphides of UG2 ore [5]. On the plant side, Mintek's AI optimisation raised a lead-zinc concentrator's throughput by 0.5%, worth over US\$2 million (about R33 million) a year [21], so throughput moves of this order are both achievable and valued.

Four ideas were tested and withdrawn, and they mark where the method stops. The belt spectra did not predict QEMSCAN mineralogy once size fraction and process line were accounted for. Belt chemistry routed Bushveld PGE ore worse than the mine plan's seam label (balanced accuracy 0.79 against 0.82). The belt did not predict recovery, lime or pH better than a baseline. Plant tags never forecast concentrate silica an hour ahead better than the last laboratory assay.

5.2 Limitations

The following limitations apply: (a) no South African ore has been evaluated, and a model trained on Chilean porphyry core may not transfer, because hardness also depends on silicification, grain size and fracturing; (b) the belt result is simulated on 146 composites without time order, and without drill-hole identifiers the split is by sample, so it may be optimistic; (c) the belt camera predicts hardness and does not identify minerals, while the microscope has not been trained on chromite or

the silicates; (d) the 12 microscope test sections guided development, and the approved checkpoint cannot be regenerated exactly; (e) the advisor’s band was set on the test sections, its other thresholds are placeholders, and liberation is measured in 2D, which overstates it; and (f) latency is measured on one laptop, and the plant is a simulator. Compared with the abstract Mintek accepted, three things changed: the five Bushveld phases became three sulphides on an analogue ore, because no labelled Bushveld images existed; the QEMSCAN “teacher” became the pilot’s truth-laboratory stream; and the fallback was reversed from holding the last setpoint to applying the conservative setting.

6 Business model and financial case

6.1 Value proposition and business model

Our value proposition is simple: the plant gets a reading of its ore before the mill, with a stated bound, and after the pilot it pays only for value that an independent party has measured. The buyer is the concentrator manager or general manager for metallurgy at a South African PGM producer, who owns recovery, concentrate quality and mill throughput; producers such as Valterra, Implats, Sibanye-Stillwater, Northam and Tharisa are market segments, not customers. Figure 11 illustrates how money and value flow, highlighting the two kinds of payment: the pilot is paid at cost, and only the later fee depends on measured value. The Technology Innovation Agency (TIA), the national agency that funds technology development [22], is the proposed bridge funder.

The revenue model has four streams: a paid laboratory pilot; a monthly licence for the laboratory software; recovery of the belt pilot’s cost; and, after a successful pilot, at most 20% of the measured net value per concentrator per year. The model suits Mintek for four reasons, each drawn from its own published work [21]: (i) it creates more truth-laboratory work, because the models need QEMSCAN composites and Bond tests to retrain and audit them; (ii) it is a feed-forward input that Mintek’s process control can use, as its AI optimisation already does with process data; (iii) Mintek has already proven value-linked pricing, since its benefit-sharing model with Hindustan Zinc in India has generated over R35 million since 2021; and (iv) it has a route beyond South Africa, because Mintek’s products and services are used in over 60 countries, its instruments group operates in 42, and it works through distributors including Molycop and Mercantil. The timing also favours a pilot now: South African PGM output fell 9% in 2025 [4], while Valterra’s realised basket price rose from R32,611 per ounce in 2025 to R45,993 in the first half of 2026 [6, 23].

Table 4 compares the options a plant can buy today. The system replaces none of them; it adds an ore-side input before the mill and uses their results as its truth.

6.2 Pricing, and what the money pays for

Table 5 sets out the price of each stage and, more importantly, what it pays for beyond equipment and software: sampling, laboratory truth, maintenance, training, security review and independent verification. TIA bridge-funds the laboratory pilot; the host producer pays the belt pilot at cost; and Mintek’s laboratory work is billed to the pilot budget.

Table 4: What plants use today, and where this system differs.

Option	What it gives	Where this differs
Cross-belt neutron analysers	Bulk elements, minute by minute	Hardness, not elements; isotope sources need a SAHPRA licence [24], this needs none
Slurry analysers	Grade and Cr_2O_3 in slurry, after milling	Reads the ore before the mill; their output is an input
Automated mineralogy (QEMSCAN)	Phases and liberation, in days	Predicts between truth results; its composites retrain the models
APC (e.g. MillStar, FloatStar)	Stable, optimised circuits from process data	A partner: an ore-side input inside APC limits

6.3 From throughput to recovered metal

The belt result is a throughput gain, and revenue comes from recovered metal, so the two must be bridged. For a plant whose mill limits production, with unchanged head grade, the extra metal recovered is approximately

$$\frac{\Delta M}{M} \approx \frac{\Delta T}{T} + \frac{\Delta R}{R}, \quad (3)$$

where T is throughput and R is recovery. Grinding faster can cost recovery: the modelled flotation cost is up to 0.25 percentage points on the steep part of the recovery curve and none on the plateau (Section 4.3), which at an 87% recovery [5] is $\Delta R/R = -0.29\%$. Where the plant is limited by ore supply instead, extra throughput is worth nothing. Table 6 turns these cases into value at Zondereinde’s output.

6.4 Financial methodology

The following assumptions apply: (a) annual output is that of Northam’s Zondereinde mine, 333,050 equivalent refined 4E ounces in F2026 [26]; (b) prices are Valterra’s realised basket prices of R32,611 per ounce (2025) and R45,993 (first half of 2026), with cash operating costs of R19,488 and R20,677 per ounce [6, 23], used as proxies; (c) costs are the ranges in Table 5; and (d) the pilot equipment has a 5-year life at a 10% discount rate. The annual value of an uplift u in recovered metal is

$$V = u \times Q \times m, \quad (4)$$

where Q is annual output in ounces and m the margin per ounce. For $u = 1\%$ and $m = \text{price less cash cost}$,

$$V = 0.01 \times 333,050 \times (\text{R}13,123 \text{ to } \text{R}25,316), \quad (5)$$

which is equivalent to R43.7–84.3 million a year (R108.6–153.2 million in revenue before costs). Deducting the full average cash cost is conservative, because extra ounces from an existing mill cost less than the average ounce. The installed cost is spread over the equipment’s life with the capital recovery factor

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} = 0.2638 \quad (r = 10\%, n = 5), \quad (6)$$

so the annualised cost is

$$C = \text{CRF} \times C_{\text{installed}} + C_{\text{running}}, \quad (7)$$

which is equivalent to R3.8 million a year (low), R11.4 million (middle) and R22.8 million (high), or R25.4 million for the

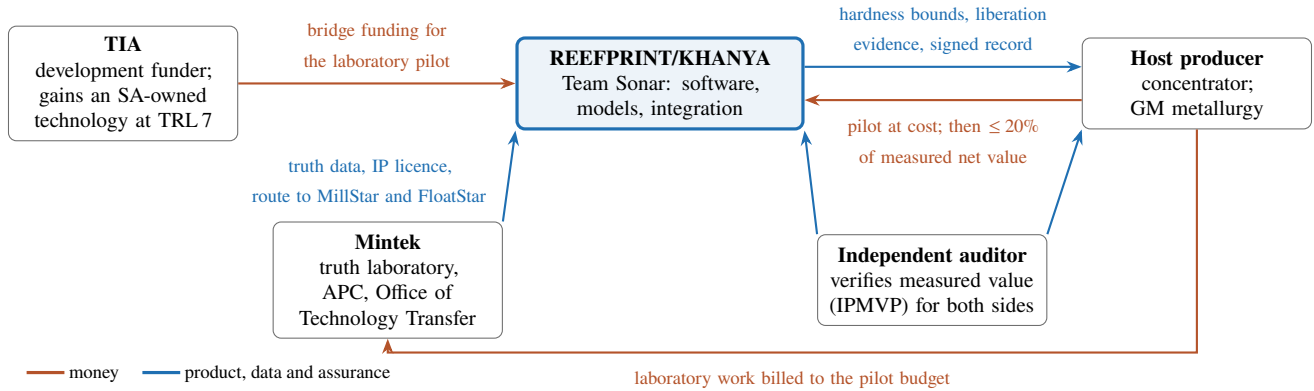


Figure 11: The business model. The producer pays for the pilot at cost, as an at-risk development investment; after the pilot it pays a capped share of value that an independent party has measured, so no measured value means no fee. Mintek earns from the laboratory work the models need and licenses the IP through its Office of Technology Transfer (MOTT); TIA’s bridge funding de-risks the laboratory pilot.

Table 5: Pricing by stage. Prices are the team’s hypotheses (H) or planning assumptions (A) until vendors and laboratories quote. Ranges first set in US\$ are converted at R16.4 per US\$, the January–September 2026 average [25].

Stage	Price	What it pays for	Who pays	Basis
1. Laboratory pilot (KHANYA), 3 months	About R210,000	Team time to retrain the microscope on QEMSCAN-labelled sections of the host’s ore; an independent test set; a validation report against QEMSCAN; workbench installation; training of laboratory staff	TIA bridge funding	H
2. Laboratory licence	About R12,000 per site per month, plus a deployment fee	Software updates, model audits, support and security patches	Host laboratory	H
3. Belt pilot, one concentrator, 9 months	R8.2–49.2 million installed; R1.6–9.8 million a year to run	<i>Installed:</i> infrared and visible cameras, enclosure and window, lighting, calibration references, belt-speed encoder, edge computer with UPS, structural and electrical work, OPC UA and historian integration, an OT security assessment, commissioning. <i>Running:</i> sampling, assays, Bond tests, QEMSCAN composites, maintenance and spares, support, training, independent measurement and verification	Host producer, at cost and at risk; Mintek’s laboratory work billed to the pilot	A
4. Contingency	20% of installed cost (R1.6–9.8 million)	Released only against quotes at gate G1	Funder of each stage	A
5. After the pilot	At most 20% of measured net value per concentrator per year	Models, audits, support and upgrades; zero if no value is measured	Producer	H

Table 6: From throughput to value at Zondereinde’s output. Net value deducts Valterra’s cash operating cost per ounce (2025 and H1 2026 prices). Simulated gains, not measurements.

Case	More metal	Net value (R M/yr)	Fee cap 20% (R M/yr)
Simulated gain held in envelope; recovery on plateau	1.75%	77–148	15–30
Same, on the steep part of the recovery curve	1.46%	64–123	13–25
A quarter of the simulated gain	0.44%	19–37	4–7
Plant limited by ore supply	0	0	0

high case with contingency. The break-even uplift is then $u_{BE} = C/(Q \times m)$: 0.05% in the best case (low cost, 2026 margin) and 0.52% in the worst (high cost, 2025 margin), or 0.58% with contingency. Figure 12 illustrates the break-even, highlighting that even the worst case needs less than a third of the simulated gain.

Annualising over five years describes a pilot that succeeds. If it fails, the producer’s loss is what it has spent: R8.5–50.8 million if it stops at G1 (month 2), R8.9–53.3 million at G2 (month 5), and R9.4–56.6 million at G3 (month 9), or up to R66.4 million if the whole contingency is used. No salvage value is assumed. A 30% lower basket price would remove

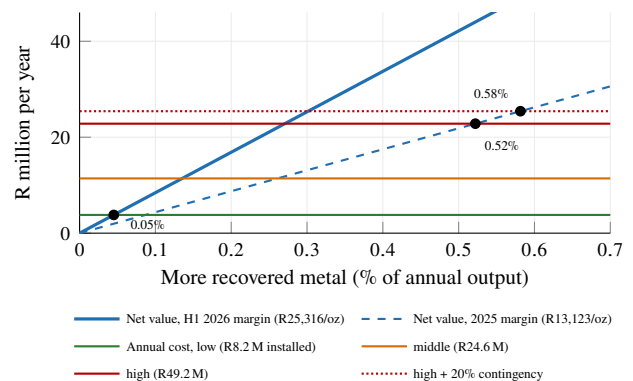


Figure 12: Break-even for one belt at Zondereinde’s output: annual net value of extra recovered metal (blue) against the annualised pilot cost (horizontal lines). Dots mark the best case, the worst case and the worst case with contingency.

most of the 2025 margin, which is why the fee is a share of measured net value rather than a fixed charge.

6.5 Risks, contingency and fallbacks

Figure 13 and Table 7 set out the main risks of using the product, as the team assesses them. Each has a mitigation that acts before it happens and a fallback that limits the damage if it

Impact	High	R4	R1 R6	
	Medium	R10	R5 R7 R9	R2
	Low		R8	R3
		Low	Medium	High
		Likelihood		

Figure 13: Risk matrix (team’s assessment before mitigation). Risks R1–R10 are described in Table 7.

does. Three fallbacks run through all of them: the plant always has its conservative setting and its existing workflow; each pilot phase ends at a gate where spending can stop; and the fee is zero if no value is measured.

7 Implementation and support

The pilot depends on people more than on hardware. Appendix C gives the detailed plans; this section summarises them.

Support (Table 11). The host site provides a sponsor, a process metallurgist, an analytical chemist, a process control engineer, an instrumentation technician and an OT security officer; Mintek provides a mineralogist and MOTT; Team Sonar provides three full-time roles.

Maintenance (Table 13). Most checks are automatic, and the system refuses rather than guesses when a check fails, so poor maintenance shows up as more conservative settings, not as wrong advice.

Compliance (Table 12). The Mine Health and Safety Act requires a risk assessment with the health and safety committee and training before significant changes to plant or equipment [27]; POPIA governs approvers’ names in the record [28]; Mining Charter clause 2.3.1 counts skills and R&D spend [29]; IP from public funding falls under the IPR Act [30]; Mintek and TIA spend under the PFMA [31]; and the camera needs no radiation licence. Before installation, the health and safety committee, the management-of-change review, the IT/OT change board and the capital committee sign off.

Independent fact-checkers to outsource: (a) a SANAS-accredited laboratory for Bond tests and assays [32]; (b) Mintek’s QEMSCAN for mineralogy; (c) an independent auditor for the value on which fees are charged, using IPMPV [33]; (d) an IEC 62443 assessor for operational-technology security [34]; (e) an IP attorney through MOTT; and (f) a university reviewer for the model audit.

Training (Table 14). Each group receives a short module before its phase starts, aligned with the site’s workplace skills plan, which mines report to the Mining Qualifications Authority [35].

8 Roadmap after funding approval

Figure 14 illustrates the plan from the day funding is approved, highlighting that each phase ends at a gate the partners agree in advance. If a gate fails, the pilot stops, and the spending to that point is the maximum exposure (Section 6.4). Table 8 lists the evidence each gate must show. After G3 the system is coupled to the pilot site’s process control within its limits, a second site on another reef tests transfer by month 18, and

the South African rollout then proceeds producer by producer, with Mintek’s network offering a route to Zimbabwe’s Great Dyke and the Copperbelt from month 24.

9 Conclusions and recommendation

The technical verdict is promising but unproven on South African ore. On public data, a belt hyperspectral model predicted grinding hardness well enough to steer a mill (simulated +1.7–1.9% throughput, no more overloads), and a microscope model identified the three main base-metal sulphides of UG2 ore without unsafe advice, although its test sections also guided development. The financial verdict does not depend on large effects: even after cash costs and in the worst cost case, one belt breaks even at about half a per cent more recovered metal. We therefore recommend that TIA, Mintek and one PGM producer approve a nine-month pilot, one belt and one truth laboratory, with gates at months 2, 5 and 9. The single most important uncertainty, whether the hardness model transfers to Bushveld ore, is answered by month 5.

CRedit authorship contribution statement. **Lethabo Hoaeane:** Conceptualisation, domain lead, belt hyperspectral modelling, belt monitor and decision application, economics, presentation. **Sibusiso Khumalo:** Microscope segmentation model, accuracy and decision-level evaluation, advisor and safety gates, OPC UA simulation, workbench review and verification, technical report.

Declaration of competing interest. The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Use of AI tools. Code and drafting were assisted by AI coding assistants. Every reported number was computed by code in the project repository and checked by the authors.

Data and code availability. HIDSAG is available under CC0 (doi:10.6084/m9.figshare.c.5983921) [9]; LumenStone under research-use terms [10]; the Bushveld assays under CC BY 4.0 (doi:10.17632/dc8jcnbcvk.1) [11]. Code, checkpoint hashes and every results file behind the numbers in this report: <https://github.com/Sibusiso-K/KHANYA> (Appendix D). A static demonstration of the belt monitor runs at <https://lethabomh14-reefprint.static.hf.space>.

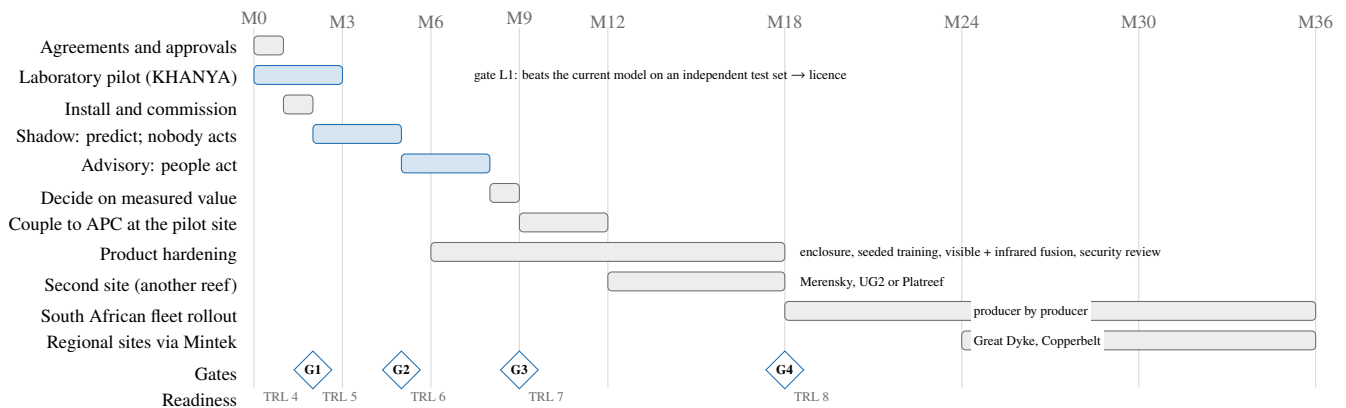
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Table 7: Risk register: mitigation before the event, and the contingency or fallback if it happens.

#	Risk	Mitigation (before)	Contingency or fallback (if it happens)
R1	The hardness model does not transfer to Bushveld ore	Laboratory pilot on the host's ore first; shadow phase compares bounds with site Bond tests (G2)	Retrain on site data, or stop at month 5
R2	Belt conditions: dust, moisture, vibration, lamp ageing, wavelength drift	Purge air, calibration references and a calibration gate at commissioning (G1)	Flagged parcels get the conservative setting; the plant runs as it does today
R3	Power outages and load-shedding	Runs offline on an edge computer with a UPS	After a restart, the conservative setting holds until checks pass
R4	Cyber attack or an unsafe write to the plant	Read-only advisory tags; no write path to safety systems; IEC 62443 zones; signed records	Disconnect; the plant continues on its existing control system
R5	Operators ignore, over-trust or cannot keep up with the advice	Deadband and shift-level approvals; training before the advisory phase; named roles	Escalation; the conservative setting applies when nobody responds
R6	No measurable gain	On/off operating blocks with confidence intervals; independent measurement and verification	Stop at G3; no fee is due
R7	Cost overrun or a weaker rand	Quotes before G1; 20% contingency; funding in tranches at gates	Defer the belt pilot and continue the laboratory pilot
R8	A fall in PGM prices	The fee is a share of measured net value	The fee falls with value automatically
R9	Funding delay	The laboratory pilot is small (about R210,000) and can start first	Laboratory pilot only; the belt pilot starts when funded
R10	Privacy or labour objections to the decision record	POPIA purpose, retention and non-disciplinary use agreed first	Record approval roles instead of names

**Figure 14:** Roadmap from funding approval (M0) to month 36. Shaded bars are the nine-month pilot's core phases; diamonds are go/no-go gates (Table 8). Technology readiness levels (TRL) are the team's self-assessment. Months beyond M9 depend on G3.**Table 8:** Milestones and go/no-go tests after funding approval.

Gate	Month	Evidence to pass	If it fails
–	1	Agreements signed (data, POPIA, OT access, IP); envelope approved	Do not install
G1	2	References and calibration pass; no write path to any safety system	Fix before shadow
L1	3	Microscope retrained on host ore beats the current model on an independent test set; verify band re-set	Stay at research level
G2	5	Bound coverage holds against site Bond tests; OOD rate and proposal rate acceptable	Retrain or stop
G3	9	Measured gain above break-even, with a confidence interval, verified independently	Stop; no fee
G4	18	A second site passes G2 within budget	Stay single-site

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A Glossary

APC	Advanced process control, e.g. Mintek’s MillStar and FloatStar
Bond work index (W_i)	Energy needed to grind ore, in kWh/t; harder ore has a higher W_i
Bound	Here, a hardness the true value should stay below 9 times in 10 (one-sided 90% conformance bound)
Conservative setting	The feed rate set for hard ore (the training data’s 90th-percentile hardness); used whenever the system is unsure
Deadband	The smallest change worth showing the operator
Envelope	The range of feed rates the site’s metallurgist allows (demonstration: 85–110% of design)
Free / locked	A grain at least / less than half payload by area
IoU	Intersection over union: overlap of predicted and expert areas, 0 to 1
IPMVP	International Performance Measurement and Verification Protocol
MOTT	Mintek’s Office of Technology Transfer
OOD	Out of distribution: unlike anything in training
OPC UA	The industrial data-exchange standard IEC 62541
Parcel	30 s of belt in the pilot design; one composite scan in the replay
PGM, 4E	Platinum-group metals; 4E = Pt, Pd, Rh and Au
QEMSCAN	Automated scanning-electron mineralogy
SWIR, VNIR	Short-wave and visible–near infrared light
TIA	Technology Innovation Agency
TRL	Technology readiness level, 1 to 9

B Microscope detail

Tables 9 and 10 (final page) give per-class results for the approved checkpoint and the retrained candidate, and the pixel confusion matrix.

Per-section mean IoU (test_01 to test_12): 0.413, 0.728, 0.659, 0.466, 0.472, 0.340, 0.453, 0.504, 0.375, 0.378, 0.377, 0.443. Chalcopyrite and pentlandite are confused with each other and with pyrrhotite; that, not magnetite, limits the payload measures.

C Implementation plans

Tables 11–14 (final page) give the support, compliance, maintenance and training plans.

D Software verification and reproducibility

Tests. The microscope workbench has 181 unit-test functions and 7 browser tests, with four continuous-integration checks passing; the belt software has 328 unit-test functions. Static security analysis reported no issues after a dependency upgrade.

Checkpoints (SHA-256). Approved:

de7135a96541a46dc0981a991cb954186c7cd669ea1c1b33914d1929ae9b1357.

Candidate:

42646cfafbeac5398386dba17f7ad3531ffcea572e6342cc4fcb42ed0b44b3b.

Reproducibility. The training patch sampler was unseeded, so a second run of the approved recipe scored 0.454; it is now seeded by seed, epoch and index, with a test that two runs draw identical patches (commit adeb1ee). The approved checkpoint itself cannot be regenerated exactly. The image-analysis runtime is pinned (requirements-lock.txt); pinning the training environment and a seeded re-run are pilot tasks.

Table 9: Microscope per-class results, approved checkpoint against the retrained candidate, 12 S2 test sections (full-section inference).

Class	Approved de7135a9			Candidate
	IoU	Recall	Precision	IoU (precision)
Resin	0.871	0.920	0.942	0.848 (0.960)
Chalcopryrite	0.576	0.716	0.746	0.800 (0.937)
Magnetite	0.000	0.000	–	0.248 (0.255)
Pyrrhotite	0.870	0.934	0.926	0.818 (0.950)
Pentlandite	0.547	0.742	0.675	0.446 (0.503)
Mean	0.573			0.632

Table 10: Pixel confusion for the approved checkpoint (% of each true class), 103.8 million test pixels.

True \ predicted	Resin	Ccp	Mag	Po	Pn
Resin	92.0	0.8	0.0	5.7	1.5
Chalcopryrite	1.6	71.6	0.0	3.1	23.7
Magnetite	78.3	0.1	0.0	16.8	4.8
Pyrrhotite	1.1	1.6	0.0	93.4	3.9
Pentlandite	0.7	0.8	0.0	24.3	74.2

Table 11: Support needed from the host site, Mintek and the funders. Time commitments are planning assumptions.

Role	Why the pilot needs them	When (month)	Commitment
Plant manager or GM metallurgy	Sponsor; owns the budget, approves the envelope, decides at each gate	0–9	Gate meetings
Process metallurgist	Sets the envelope, deadband and conservative setting; reviews the record	0–9	0.3 full-time
Mineralogist (Mintek)	QEMSCAN truth; labels sections for retraining and the independent test set	0–9	0.2 full-time
Analytical chemist	Assays and Cr ₂ O ₃ ; quality control with blanks, duplicates and reference materials	2–9	Per sampling plan
Process control engineer	OPC UA tags, historian access, later coupling to APC	1–2, then 9+	As needed
Instrumentation engineer and technician	Installation, enclosure, references, lamps	1–9	0.3 full-time
OT and IT security officer	Network zone, read-only interface, backups	0–2	0.1 full-time
Health and safety representatives; information officer	Risk assessment before installation (MHSA); POPIA and the record policy	0–1	Consultation
Mintek MOTT; TIA portfolio manager	IP agreement; funding tranches at gates	0, gates	As needed
Team Sonar	Models, software, integration, analysis and reporting	0–9	Three full-time roles

Table 12: Laws, policies and standards that apply to the pilot, and how the design complies.

Law, policy or standard	What it asks of this project	How the design complies
Mine Health and Safety Act 29 of 1996 [27]	Assess risks of new equipment, consulting the health and safety committee (s. 11, s. 34); train employees before significant changes to plant or equipment, and keep records (s. 10)	Installed under the site's risk assessment and lockout; advisory only; operators trained before the advisory phase
Protection of Personal Information Act 4 of 2013 [28]	Lawful, limited processing of personal information, such as approvers' names	Purpose, access, retention and non-disciplinary use agreed with the information officer before the shadow phase
Mining Charter 2018, clause 2.3.1 [29]	Invest 5% of the leviable amount in essential skills development, including R&D in processing and technology efficiency	Pilot training and R&D may count toward it; clause 2.3 was not among the clauses set aside by the High Court in 2021 [36]
IPR from Publicly Financed R&D Act 51 of 2008 [30]	IP from publicly financed R&D is identified, protected and commercialised for South Africa (NIPMO)	IP managed by MOTT under the hackathon agreement, with invention credits [8]
Public Finance Management Act 1 of 1999 [31]	Mintek (Schedule 3B) and TIA (Schedule 3A) spend public money under PFMA controls	Funding released in tranches tied to gates G1–G3, with auditable deliverables
Hazardous Substances Act 15 of 1973 (SAHPRA) [24]	Licences for radioactive sources (Group IV)	Not applicable: optical sensing only
IEC 62443 [34]; OPC UA [17]	Zones and conduits for industrial cyber security; secure data exchange	Read-only advisory tags; no write path to safety systems; IT/OT sign-off
ISO/IEC 17025, accredited by SANAS [32]	Demonstrated competence of testing laboratories	Bond tests and assays used as truth come from accredited laboratories
IPMVP (EVO) [33]	A recognised protocol for measuring and verifying savings	Fees invoiced only on value measured under an agreed plan by an independent party

Table 13: Proposed maintenance plan; the site sets the final intervals.

Task	Interval	Owner
Reference checks; refuse if a feature shifts over half a band	Every reference cycle (automatic)	System
Window cleaning and purge-air check	Each shift, or on alarm	Instrument technician
Lamp replacement	Lamp-life schedule	Instrument technician
Enclosure temperature and dew point	Continuous alarm	System; electrical
Microscope colour and illumination check	Every image; daily	Laboratory technician
Model audit against Bond tests and QEMSCAN	Monthly in the pilot	Metallurgist, mineralogist
Retraining	Only through a validation gate	Team Sonar
Security patches; backup restore test	Monthly; quarterly	Team Sonar, IT/OT

Table 14: Proposed training plan. Records are kept as the MHSA requires.

Who	What they learn	Format; when
Control-room operators, supervisors	Reading a proposal; approve, modify, reject or escalate; the deadband; what the fallback does	Half a day plus supervised shifts; before month 5
Metallurgists	Setting the envelope and deadband; reading bounds and liberation; reviewing the record	One day; month 1
Laboratory staff	Workbench, colour calibration, grain audit, comparison with QEMSCAN	One day; months 0–3
Instrument technicians	Camera, enclosure, references, lamps, lockout	One day with the vendor; commissioning
IT and OT staff	Network zones, backups, patching, incident response	Half a day; before installation
Managers and funders	Reading the measured-value report and the gates	One hour; each gate